

24. (2 points)

An insurance company wants to allocate capital to a newly purchased portfolio based on the portfolio's aggregate loss distribution.

The portfolio contains the following two lines of business:

- Commercial Property subject to both earthquake and hurricane risks
- Excess Casualty with a large policy limit

The following risk measures are under consideration:

- Standard deviation
- TV@R
- TV@R with transformed probabilities

a. (1.5 points)

Evaluate the appropriateness of these three risk measures for this portfolio and recommend which measure the company should adopt.

b. (0.5 point)

Identify one additional risk measure that would be suitable for this capital allocation and briefly describe its appropriateness for this portfolio.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 24	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: C3: Demonstrate the properties of various risk measures and their limitations; and C4: Describe how risk measures and risk modeling, including allocation, can affect strategic management.
SAMPLE ANSWERS	
Part a: 1.5 points	
<i>Sample Answer 1</i> • The portfolio is subject to large losses. • Standard Deviation treats the favorable deviations the same as the unfavorable deviations. • TVaR is linear in the tail meaning it does not treat a loss 2× as large as more than 2× as bad. • TVaR on transformed probabilities DOES treat a loss 2× as large as more than 2× as bad. This is the better statistic to use. <i>Sample Answer 2</i> (1) SD is not appropriate as it treats favorable outcomes the same as unfavorable, with risk capital requirements. We are solely interested in unfavorable outcomes. Also, [SD] penalizes large deviations from the mean and these lines will have skewed losses. (2) TVaR measures the average loss above VaR, but treats all losses in the tail linearly. There is high potential for skewed losses for these LOB, so a measure which treats losses in the tail linearly is not appropriate (although better than SD). (3) WTVaR places more weight on the highly unfavorable results. This is the best measure for the skewed loss distributions. It will recognize that a loss 2 times as large as more than double the impact. Select WTVaR. <i>Sample Answer 3</i> • Standard deviation includes both negative and positive outcomes. Since we want to focus on the negative, it is not appropriate. • TVaR would be more appropriate, but since it is linear in the tail, and these lines have potentially large tail, it is not preferred. • TVaR transformed prob is the most appropriate since it treats a loss twice as big as more than twice as bad. Recommended.	
Part b: 0.5 point	
<i>Sample Answer 1</i> EPD on the transformed probabilities is a tail-measure that would also address market attitudes toward risk, which is important for a book with higher likelihood in the right tail. <i>Sample Answer 2</i> Value of put option. This would take into account the market value to protect against our extreme event. So risk measure is proportional to market value which is what we want.	

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Sample Answer 3

Exponential Moment:

- 1 It considers ALL losses in the distn, not just the tails. This is good since company may suffer medium sized losses not captured by TVaR.
- 2 It still reflects skewness of the distn, unlike SD, so it works well for this portfolio.

EXAMINER'S REPORT

Candidates were expected to know the definition and key properties of each risk measure and its applicability to capital allocation for the given risk portfolio. Candidates generally scored well on both parts of the question. Incorrect answers typically failed to include explanations or were not responsive to the question asked.

Part a

The most common error was a failure to differentiate between TVaR and TVaR with transformed probabilities. Most correct answers noted that TVaR reflected a linear risk preference, which is inconsistent with risk aversion. A few candidates failed to indicate which measure they would recommend.

Part b

The most common error was selection of a synonym of a measure already listed in part a. Common examples were Weighted TVaR (same as TVaR with transformed probabilities) or Conditional Tail Expectation (same as TVaR). A few candidates gave answers like Wang Transform or Copula with Heavy Right Tails (HRT). These are distributions, not risk measures, which may be components of a correct answer.